

Railway Reservation System

```
CREATE TABLE trains_22phcs004a (
```

```
    TrainID INT PRIMARY KEY,
```

```
    TrainName VARCHAR(255),
```

```
    Source VARCHAR(255),
```

```
    Destination VARCHAR(255),
```

```
    TotalSeats INT
```

```
);
```

```
describe Trains_22phcs004a;
```

```
Name
```

```
Null?
```

```
Type
```

```
-----
```

```
TRAINID
```

```
NOT NULL NUMBER(38)
```

TRAINNAME	VARCHAR2 (255)
SOURCE	VARCHAR2 (255)
DESTINATION	VARCHAR2 (255)
TOTALSEATS	NUMBER (38)

```
CREATE TABLE Passengers_22phcs004a (
  PassengerID INT PRIMARY KEY,
  FirstName VARCHAR(255),
  LastName VARCHAR(255),
  Age INT,
  Gender CHAR(1)
);
```

```
describe passengers_22phcs004a;
```

Name	Null?	Type
-----	-----	-----

PASSENGERID	NOT NULL NUMBER (38)
FIRSTNAME	VARCHAR2 (255)
LASTNAME	VARCHAR2 (255)
AGE	NUMBER (38)
GENDER	CHAR (1)

```
CREATE TABLE Schedules_22phcs004a (  
    ScheduleID INT PRIMARY KEY,  
    TrainID INT,  
    DepartureTime varchar(20),  
    ArrivalTime varchar(20),  
    FOREIGN KEY (TrainID) REFERENCES Trains_22phcs004a (TrainID)  
);  
  
describe Schedules_22phcs004a;
```

Name	Null?	Type
-----	-----	-----
SCHEDULEID	NOT NULL	NUMBER (38)
TRAINID		NUMBER (38)
DEPARTURETIME		VARCHAR2 (20)
ARRIVALTIME		VARCHAR2 (20)

```

CREATE TABLE Bookings_22phcs004a (
    BookingID INT PRIMARY KEY,
    PassengerID INT,
    ScheduleID INT,
    SeatNumber VARCHAR(10),
    BookingDay varchar(20),
    FOREIGN KEY (ScheduleID) REFERENCES Schedules_22phcs004a (ScheduleID)

```

);

describe Bookings_22phcs004a;

Name	Null?	Type

BOOKINGID	NOT NULL	NUMBER (38)
PASSENGERID		NUMBER (38)
SCHEDULEID		NUMBER (38)
SEATNUMBER		VARCHAR2 (10)
BOOKINGDAY		VARCHAR2 (20)

```
create table tickets_22phcs004a (  
tickets_id varchar(20),  
bookingid int,  
ticket_number varchar(20),  
ticket_prize varchar(20)  
);
```

describe tickets_22phcs004a;

Name	Null?	Type

TICKETS_ID		VARCHAR2 (20)
BOOKINGID		NUMBER (38)
TICKET_NUMBER		VARCHAR2 (20)
TICKET_PRIZE		VARCHAR2 (20)

insert into trains_22phcs004a (TrainID,TrainName,Source, Destination,TotalSeats) values (1472,'jodpur express','adur road','jodpur junction',200);

1 row created.

insert into trains_22phcs004a(TrainID,TrainName,Source, Destination,TotalSeats) values(1473,'prasanthi express','kadiri road','vijayawada junction',2000);

1 row created.

```
insert into trains_22phcs004a( TrainID,TrainName,Source, Destination,TotalSeats )  
values(1474,'venkatadhari express','vijayawada road','hyderabd junction',3500);
```

1 row created.

```
insert into trains_22phcs004a( TrainID,TrainName,Source, Destination,TotalSeats ) values(1475,'purulia  
express','purulia road','adra junction',1000);
```

1 row created.

```
insert into trains_22phcs004a( TrainID,TrainName,Source, Destination,TotalSeats ) values(1476,'agartala  
express','agartala road','anand junction',1500);
```

1 row created.

set linesize 150

COLUMN TrainID FORMAT 99999

COLUMN TrainName FORMAT A20

COLUMN Source FORMAT A30

COLUMN Destination FORMAT A25

COLUMN TotalSeats FORMAT 99999

select * from trains_22phcs004a;

TRAINID	TRAINNAME	SOURCE	DESTINATION	TOTALSEATS
1472	jodpur express	adur road	jodpur junction	200
1473	prasanthi express	kadiri road	vijayawada junction	2000

1474	venkatadhari express	vijayawada road	hyderabd junction	3500
1475	purulia express	purulia road	adra junction	1000
1476	agartala express	agartala road	anand junction	1500

```
insert into passengers_22phcs004a (PassengerID , FirstName, LastName, Age, Gender) values  
(1,'nethra','gowda',20,'f');
```

1 row created.

```
insert into passengers_22phcs004a (PassengerID , FirstName, LastName, Age, Gender)  
values(2,'bommi','moksha',22,'f');
```

1 row created.

```
insert into passengers_22phcs004a (PassengerID , FirstName, LastName, Age, Gender)
values(3,'omisha','gopalraj',21,'f');
```

1 row created.

```
insert into passengers_22phcs004a (PassengerID , FirstName, LastName, Age, Gender)
values(4,'lohith','v',19,'m');
```

1 row created.

COLUMN PassengerID FORMAT 99999

COLUMN FirstName FORMAT A30

COLUMN LastName FORMAT A40

COLUMN Age FORMAT 99999

COLUMN Gender FORMAT A10

SELECT * FROM passengers_22phcs004a;

PASSENGERID	FIRSTNAME	LASTNAME	AGE	GENDER
-------------	-----------	----------	-----	--------

1	nethra	gowda	20	f
2	bommi	moksha	22	f
3	omisha	gopalraj	21	f
4	lohith	v	19	m

insert into Schedules_22phcs004a (ScheduleID, TrainID , DepartureTime , ArrivalTime) values
(101,1472,'10:30','16:30');

1 row created.

```
insert into Schedules_22phcs004a ( ScheduleID, TrainID , DepartureTime , ArrivalTime ) values  
(102,1473,'11:30','12:27');
```

1 row created.

```
insert into Schedules_22phcs004a ( ScheduleID, TrainID , DepartureTime , ArrivalTime ) values  
(103,1474,'21:00','09:50');
```

1 row created.

```
insert into Schedules_22phcs004a ( ScheduleID, TrainID , DepartureTime , ArrivalTime ) values  
(104,1475,'14:30','17:10');
```

1 row created.

```
insert into Schedules_22phcs004a ( ScheduleID, TrainID , DepartureTime , ArrivalTime ) values  
(105,1476,'05:30','09:25');
```

1 row created.

COLUMN ScheduleID FORMAT 99999

COLUMN TrainID FORMAT 99999

COLUMN DepartureTime FORMAT A30

COLUMN ArrivalTime FORMAT 99999

SELECT * FROM Schedules_22phcs004a;

SCHEDULEID	TRAINID	DEPARTURETIME	ARRIVALTIME
------------	---------	---------------	-------------

101	1472	10:30	16:30
-----	------	-------	-------

102	1473	11:30	12:27
-----	------	-------	-------

103	1474	21:00	09:50
-----	------	-------	-------

104	1475	14:30	17:10
-----	------	-------	-------

105	1476	05:30	09:25
-----	------	-------	-------

insert into Bookings_22phcs004a (BookingID ,PassengerID ,ScheduleID ,SeatNumber ,BookingDay) values
(121,1,101,22,'mon');

1 row created.

```
insert into Bookings_22phcs004a (BookingID ,PassengerID ,ScheduleID ,SeatNumber ,BookingDay ) values  
(122,2,102,21,'fri');
```

1 row created.

```
insert into Bookings_22phcs004a (BookingID ,PassengerID ,ScheduleID ,SeatNumber ,BookingDay ) values  
(123,3,103,08,'wed');
```

1 row created.

```
insert into Bookings_22phcs004a (BookingID ,PassengerID ,ScheduleID ,SeatNumber ,BookingDay ) values  
(124,4,104,19,'sun');
```

1 row created.

```
insert into Bookings_22phcs004a (BookingID ,PassengerID ,ScheduleID ,SeatNumber ,BookingDay ) values  
(125,5,105,30,'mon');
```

1 row created.

```
COLUMN BookingID FORMAT 99999
```

```
COLUMN PassengerID FORMAT 99999
```

```
COLUMN ScheduleID FORMAT 99999
```

```
COLUMN SeatNumber FORMAT 99999
```

```
COLUMN BookingDay FORMAT A20
```

```
SELECT * FROM Bookings_22phcs004a;
```


BOOKINGID PASSENGERID SCHEDULEID SEATNUMBER BOOKINGDAY

121	1	101 22	mon
122	2	102 21	fri
123	3	103 8	wed
124	4	104 19	sun
125	5	105 30	mon

insert into tickets_22phcs004a values ('a1',121,'s001','2000');

1 row created.

insert into tickets_22phcs004a values ('a2',122,'s002','2500');

1 row created.

```
insert into tickets_22phcs004a values ('a3',123,'s003','3000');
```

1 row created.

```
insert into tickets_22phcs004a values ('a4',126,'s004','1000');
```

1 row created.

```
insert into tickets_22phcs004a values ('a5',127,'s005','2200');
```

```
select * from tickets_22phcs004a ;
```

TICKETS_ID	BOOKINGID	TICKET_NUMBER	TICKET_PRIZE
------------	-----------	---------------	--------------

a1	121 s001	2000
----	----------	------

a2	122 s002	2500
----	----------	------

a3	123 s003	3000
----	----------	------

a4	126 s004	1000
----	----------	------

a5	127 s005	2200
----	----------	------

To write SQL queries to implement the following concepts:

1. Select queries with WHERE clause

[The **WHERE clause** is used to filter records]

Operator	Description
=	Equal
>	Greater than
<	Less than
>=	Greater than or equal

<=	Less than or equal
<>	Not equal. Note: In some versions of SQL this operator may be written as !=
BETWEEN	Between a certain range
LIKE	Search for a pattern
IN	To specify multiple possible values for a column

[A-1] SELECT * from trains_22phcs004a where trained >'1472';

[A-2] SELECT TrainName, Source, Destination, TotalSeats FROM trains_22phcs004a TrainID ='1472';

EMPLOYEE_S	EMPLOYEE_NAME	EMPLOYEE_ADDRESS	EMPLOYEE_DATE_OF_BIRTH
------------	---------------	------------------	------------------------

2	Alya	234 Main St	10-AUG-19
---	------	-------------	-----------

2. Select queries with sort using order by clause

[The **ORDER BY** keyword is used to sort the result-set in ascending or descending order]

[B-1] SELECT * FROM trains_22phcs004a ORDER BY TrainName ASC;

EMPLOYEE_S	EMPLOYEE_NAME	EMPLOYEE_ADDRESS	EMPLOYEE_DATE_OF_BIRTH	EMPLOYEE_GENDER	DEPT_ID
------------	---------------	------------------	------------------------	-----------------	---------

2	Alya	234 Main St	10-AUG-19	female	20
---	------	-------------	-----------	--------	----

5	Anav	567 Main St	15-JAN-20	male	50
3	Arav	345 Main St	22-MAY-18	male	30
4	Jo	456 Main St	18-FEB-21	female	40
1	Kushi	123 Main St	15-JAN-20	female	10
7	rohan	127 main st	16-JAN-20	male	10

[B-2] SELECT * FROM trains_22phcs004a ORDER BY TrainName DESC;

EMPLOYEE_S	EMPLOYEE_NAME	EMPLOYEE_ADDRESS	EMPLOYEE_DATE_OF_BIRTH	EMPLOYEE_GENDER	DEPT_ID
7	rohan	127 main st	16-JAN-20	male	10
1	Kushi	123 Main St	15-JAN-20	female	10
4	Jo	456 Main St	18-FEB-21	female	40

3	Arav	345 Main St	22-MAY-18	male	30
5	Anav	567 Main St	15-JAN-20	male	50
2	Alya	234 Main St	10-AUG-19	female	20

3. To alter the structure of the tables by adding new columns

[The **ALTER TABLE** statement is used to add, delete, or modify columns in an existing table.]

```
ALTER TABLE trains_22phcs004a
```

```
ADD AvailableSeats varchar(200);
```

Table altered.

```
DESCRIBE Trains_22phcs004a;
```

Name	Null?	Type
------	-------	------

EMPLOYEE_SSN	NOT NULL VARCHAR2(10)
EMPLOYEE_NAME	VARCHAR2(30)
EMPLOYEE_ADDRESS	VARCHAR2(200)
EMPLOYEE_DATE_OF_BIRTH	DATE
EMPLOYEE_GENDER	VARCHAR2(20)
DEPT_ID	NUMBER(38)
SALARY	NUMBER(10,2)

[The **UPDATE statement** is used to modify the existing records in a table.]

UPDATE Trains_22phcs004a

SET AvailableSeats = 05

WHERE TrainID = '1472';

1 row updated.

UPDATE Trains_22phcs004a

SET AvailableSeats = 20

WHERE TrainID = '1473';

1 row updated.

UPDATE Trains_22phcs004a

SET AvailableSeats = 12

WHERE TrainID = '1474';

1 row updated.

UPDATE Trains_22phcs004a

SET AvailableSeats = 250

WHERE TrainID = '1475';

1 row updated.

SELECT * FROM Trains_22phcs004a;

EMPLOYEE_SSN	EMPLOYEE_NAME	EMPLOYEE_ADDRESS	EMPLOYEE_DATE_OF_BIRTH
EMPLOYEE_GENDER	DEPT_ID		

SALARY

1	Kushi	123 Main St	15-JAN-20	female
---	-------	-------------	-----------	--------

10
80000

2	Alya	234 Main St	10-AUG-19	female
---	------	-------------	-----------	--------

20
90000

3	Arav	345 Main St	22-MAY-18	male
---	------	-------------	-----------	------

30
50000

4	Jo	456 Main St	18-FEB-21	female
---	----	-------------	-----------	--------

40

90000

5	Anav	567 Main St	15-JAN-20	male
---	------	-------------	-----------	------

50

100000

7	rohan	127 main st	16-JAN-20	male
---	-------	-------------	-----------	------

10

20000

set linesize 150

COLUMN employee_ssn FORMAT A10

COLUMN employee_name FORMAT A20

COLUMN employee_address FORMAT A30

COLUMN employee_date_of_birth FORMAT A25

COLUMN employee_gender FORMAT A40

COLUMN dept_id FORMAT 99999

COLUMN SALARY FORMAT 999999

select * from Trains_22phcs004a;

EMPLOYEE_S	EMPLOYEE_NAME	EMPLOYEE_ADDRESS	EMPLOYEE_DATE_OF_BIRTH
EMPLOYEE_GENDER	DEPT_ID	SALARY	

-

1	Kushi	123 Main St	15-JAN-20	female	10 80000
2	Alya	234 Main St	10-AUG-19	female	20 90000
3	Arav	345 Main St	22-MAY-18	male	30 50000
4	Jo	456 Main St	18-FEB-21	female	40 90000
5	Anav	567 Main St	15-JAN-20	male	50 100000

7	rohan	127 main st	16-JAN-20	male	10	20000
---	-------	-------------	-----------	------	----	-------

4. Select queries with inner join.

[The **INNER JOIN** keyword selects records that have matching values in both tables.]

```
select Bookings_22phcs004a.bookingday,ticket_22phcs004a.ticket_prize
from Bookings_22phcs004a Bookings_22phcs004a
inner join tickets ticket_22phcs004a on bookings_22phcs004a.bookingid=ticket_22phcs004a.bookingsid;
```

EMPLOYEE_NAME	DEPENDENT_RELATION	DEPENDENT_NAME
---------------	--------------------	----------------

Kushi	aunty	vineesha
-------	-------	----------

Alya	friend	sree
------	--------	------

Arav	father	john
------	--------	------

5. Select queries with outer join.

[left join]

[The **LEFT JOIN** keyword returns all records from the left table (table1), and the matching records from the right table (table2).]

```
select bookings_22phcs004a.bookingday,ticket_22phcs004a.ticket_prize
```

```
from Bookings_22phcs004a Bookings_22phcs004a
```

```
left join tickets ticket_22phcs004a on bookings_22phcs004a.bookingid=ticket_22phcs004a.bookingsid;
```

```
EMPLOYEE_NAME    DEPENDENT_NAME
```

```
-----
```

```
Kushi            vineesha
```

```
Alya             sree
```

```
Arav             john
```

```
Jo
```


rohan

Anav

[right join]

[The **RIGHT JOIN** keyword returns all records from the right table (table2), and the matching records from the left table (table1).]

```
select bookings_22phcs004a.bookingday,ticket_22phcs004a.ticket_prize  
from Bookings_22phcs004a Bookings_22phcs004a  
right join tickets ticket_22phcs004a on bookings_22phcs004a.bookingid=ticket_22phcs004a.bookingsid;
```

EMPLOYEE_NAME	DEPENDENT_NAME
---------------	----------------

Kushi vineesha

Alya sree

Arav john

 Kiran

[full join]

[The **FULL JOIN** keyword returns all records when there is a match in left (table1) or right (table2) table records.]

```
select bookings_22phcs004a.bookingday,ticket_22phcs004a.ticket_prize
```

```
from Bookings_22phcs004a Bookings_22phcs004a
```

```
full join tickets ticket_22phcs004a on bookings_22phcs004a.bookingid=ticket_22phcs004a.bookingsid;
```

```
EMPLOYEE_NAME      DEPENDENT_NAME
```

Kushi vineesha

Alya sree

Arav john

Jo

Anav

rohan

kiran

6. Select queries with self-join. (employee table)

[A **self join** is a regular join, but the table is joined with itself.]

```
SELECT A. TRAINNAME AS TRAINNAME1, B. TRAINNAME AS TRAINNAME2, A. TRAINID  
FROM Trains_22phcs004a A, Trains_22phcs004a B  
WHERE A. TRAINID = B. TRAINID
```

ORDER BY A. TRAINID;

EMPLOYEE_NAME1	EMPLOYEE_NAME2	DEPT_ID
----------------	----------------	---------

Kushi	Kushi	10
Alya	Alya	20
Arav	Arav	30
Jo	Jo	40
Anav	Anav	50

SELECT A. TRAINNAME AS TRAINNAME1, B. TRAINNAME AS TRAINNAME2, A. TRAINID

FROM Trains_22phcs004a A, Trains_22phcs004a B

WHERE A. TotalSeats <> B. TotalSeats

AND A. TRAINID = B. TRAINID

ORDER BY A. TRAINID;

EMPLOYEE_NAME1	EMPLOYEE_NAME2	DEPT_ID

Kushi	rohan	10
rohan	Kushi	10

7. Select queries with sub queries

SELECT TrainName , Source

FROM Trains_22phcs004a

WHERE TrainID IN (

SELECT TrainID

FROM Passengers_22phcs004a

```
WHERE PassengerID = (  
SELECT PassengerID  
FROM Trains_22phcs004a  
WHERE TotalSeats = '200'  
)  
);
```

EMPLOYEE_NAME	EMPLOYEE_ADDRESS
---------------	------------------

Kushi	123 Main St
Alya	234 Main St
Arav	345 Main St
Jo	456 Main St

Anav	567 Main St
rohan	127 main st

8. Select queries with built-in numerical functions

[A] SELECT AVG(TotalSeats) AS average_Seats FROM Trains_22phcs004a;

AVERAGE_SALARY

71666.6667

[B] SELECT

SUM(TotalSeats) AS total_Seats,

AVG(TotalSeats) AS average_Seats,

MIN(TotalSeats) AS minimum_Seats,

MAX(TotalSeats) AS maximum_Seats

FROM

Trains_22phcs004a;

TOTAL_SALARY AVERAGE_SALARY MINIMUM_SALARY MAXIMUM_SALARY

430000 71666.6667 20000 100000

9. Select queries with built-in string functions

(UPPER CASE)

select firstname,upper(firstname) as uppercasename from passengers_22phcs004a;

EMPLOYEE_NAME UPPERCASE_NAME

Kushi	KUSHI
Alya	ALYA
Arav	ARAV
Jo	JO
Anav	ANAV
rohan	ROHAN

(LOWER CASE)

**select firstname,lower(firstname) as lowercasename from
passengers_22phcs004a;**

EMPLOYEE_NAME	LOWERCASE_NAME
---------------	----------------

Kushi	kushi
Alya	alya
Arav	arav
Jo	jo
Anav	anav
rohan	rohan

10. Select queries with build-in data functions

(for date)

```
SELECT BOOKINGID , EXTRACT(DAY FROM BOOKINGDAY) AS booking_day  
FROM Bookings_22phcs004a;
```

EMPLOYEE_NAME	BIRTH_DAY
---------------	-----------

-----	-----
-------	-------

Kushi	15
-------	----

Alya	10
------	----

Arav	22
------	----

Jo	18
----	----

Anav	15
------	----

rohan	16
-------	----

11. To alter the structure of the table by changing existing columns.

Example 1: Modify a Column's Data Type

[11.1] DESCRIBE Trains_22phcs004a;

ALTER TABLE Trains_22phcs004a

MODIFY TrainName VARCHAR(250);

DESCRIBE Trains_22phcs004a;

[11.2] ALTER TABLE Trains_22phcs004a ADD new_NAME VARCHAR(5);

DESCRIBE Trains_22phcs004a;

SELECT * FROM Trains_22phcs004a;

**[11.3] UPDATE Trains_22phcs004a SET new_NAME =
TO_CHAR(TrainName);**

5 rows updated.

SELECT * FROM Trains_22phcs004a;

[11.4] ALTER TABLE Trains_22phcs004a DROP COLUMN TrainName;

Table altered.

SELECT * FROM EMPLOYEE_22PHCS004A;

**[11.5] ALTER TABLE Trains_22phcs004a RENAME COLUMN new_NAME TO
TrainName;**

Table altered.

DESCRIBE Trains_22phcs004a;

Example 2: Modify a Column's Length

ALTER TABLE Trains_22phcs004a

MODIFY TrainName VARCHAR(50);

Table altered.

DESCRIBE Trains_22phcs004a;

Example 3: Add a Default Value to a Column

ALTER TABLE Bookings_22phcs004a

MODIFY BookingDay DAY DEFAULT SYSDAY;

Table altered.

DESCRIBE Bookings_22phcs004a;

SELECT * FROM Bookings_22phcs004a;

**insert into trains_22phcs004a (TrainID,TrainName,Source, Destination,TotalSeats)
values(1479,'jodpur express','adur road','jodpur junction',200);**

1 row created.

```
SELECT * FROM trains_22phcs004a;
```

Example 4: Rename a Column

```
ALTER TABLE Trains_22phcs004a
```

```
RENAME COLUMN TrainName TO TrainsName;
```

Table altered.

```
DESCRIBE Trains_22phcs004a;
```

Example 5: Modify a Column to NOT NULL

```
ALTER TABLE Trains_22phcs004a
```


MODIFY TrainName VARCHAR(50) NOT NULL;

Table altered.

DESCRIBE Trains_22phcs004a;

**insert into trains_22phcs004a (TrainID,TrainName,Source,
Destination>TotalSeats) values (1478,'adur road','jodpur junction',300);**

Example 6: Add a column update values and Drop the column

ALTER TABLE Trains_22phcs004a

ADD Ticketcancellation VARCHAR2(200);

Table altered.

```
DESCRIBE Trains_22phcs004a;
```

```
ALTER TABLE Trains_22phcs004a
```

```
DROP COLUMN Ticketcancellation;
```

```
DESCRIBE Trains_22phcs004a;
```

[12] To change the values in column

[BEFORE CHANGING THE VALUES IN THE COLUMN]

```
SELECT * FROM Trains_22phcs004a;
```

[AFTER CHANGING THE VALUES IN THE COLUMN]

UPDATE Trains_22phcs004a

SET AvailableSeats = 05

WHERE TrainID = '1471';

SELECT * FROM Trains_22phcs004a;

[13] To enable and disable constraints

[14] To delete unwanted values.

DELETE FROM Trains_22phcs004a

```
WHERE TrainID = '1476';
```

1 row deleted.

```
SELECT * FROM Trains_22phcs004a;
```

[15] To drop table

```
DROP TABLE table_name;
```

